Function Transformations

Algebra 1 — Shifts, Dilations, and Reflections

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1. Translations (Shifts)

Equation	Transformation	Point rule
$y = f(x) + k$	Shift **up** k	$(x, y) \to (x,\ y + k)$
$y = f(x) - k$	Shift **down** k	$(x, y) \to (x,\ y - k)$
$y = f(x - h)$	Shift **right** h	$(x, y) \to (x + h,\ y)$
$y = f(x + h)$	Shift **left** h	$(x, y) \to (x - h,\ y)$

2. Dilations

Form	Condition	Effect	Point rule
$y = a\,f(x)$	$\lvert a\rvert > 1$	Vertical **stretch**	$(x, y) \to (x,\ ay)$
$y = a\,f(x)$	$0 < \lvert a\rvert < 1$	Vertical **compression**	$(x, y) \to (x,\ ay)$
$y = f(bx)$	$\lvert b\rvert > 1$	Horizontal **compression**	$(x, y) \to (\frac{x}{b},\ y)$
$y = f(bx)$	$0 < \lvert b\rvert < 1$	Horizontal **stretch**	$(x, y) \to (\frac{x}{b},\ y)$

3. Reflections

Equation	Reflection over	Point rule	Effect
$y = -f(x)$	x-axis	$(x, y) \to (x,\ -y)$	flip up/down
$y = f(-x)$	y-axis	$(x, y) \to (-x,\ y)$	flip left/right

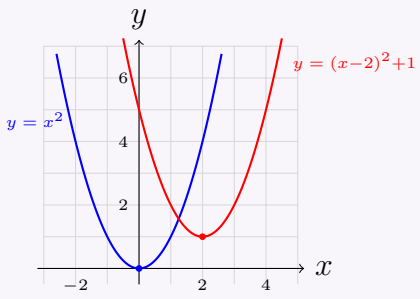
Memory rules: Outside f $\Rightarrow$ vertical change. Inside f $\Rightarrow$ horizontal change (direction **opposite** to sign).
Negating *outside* flips over x-axis. Negating *inside* flips over y-axis.

Practice Problems

Part A — Translations

Worked Example: Transform $y = x^2$ by shifting **right 2** and **up 1** to get $y = (x-2)^2+1$.

Parent $y = x^2$		**New** $y = (x-2)^2+1$	
x	y	x	y
-2	4	0	5
-1	1	1	2
0	0	2	$1 \leftarrow$ vertex
1	1	3	2
2	4	4	5



A1. Describe the transformation from $y = x^2$ to $y = x^2 + 5$.

A2. Describe the transformation from $y = x^2$ to $y = (x - 3)^2$.

A3. Describe the transformation from $y = 2^x$ to $y = 2^x - 4$.

A4. Describe the transformation from $y = 2^x$ to $y = 2^{x+1}$.

A5. Write the equation of $y = x^2$ shifted **left 2** and **down 3**.

A6. Write the equation of $y = 2^x$ shifted **right 5** and **up 2**.

A7. The point $(3, 9)$ lies on $y = x^2$. Find the corresponding point on $y = x^2 + 7$.

A8. The point $(2, 4)$ lies on $y = x^2$. Find the corresponding point on $y = (x - 1)^2$.

A9. Find the vertex of $y = (x - 4)^2 + 3$.

A10. Find the horizontal asymptote of $y = 2^x - 6$.

Part A — Multiple Choice (circle your answer)

AM1. Which point lies on $y = x^2 + 3$?

A. $(0, 0)$ **B.** $(1, 4)$ **C.** $(2, 4)$ **D.** $(-1, 2)$

AM2. The vertex of $y = (x + 4)^2$ is at:

A. $(4, 0)$ **B.** $(-4, 0)$ **C.** $(0, 4)$ **D.** $(0, -4)$

AM3. Which equation shifts $y = 2^x$ right 3 units and down 1 unit?

A. $y = 2^{x+3} - 1$ **B.** $y = 2^{x-3} + 1$ **C.** $y = 2^{x-3} - 1$ **D.** $y = 2^{x+3} + 1$

AM4. The horizontal asymptote of $y = 2^{x-2} + 5$ is:

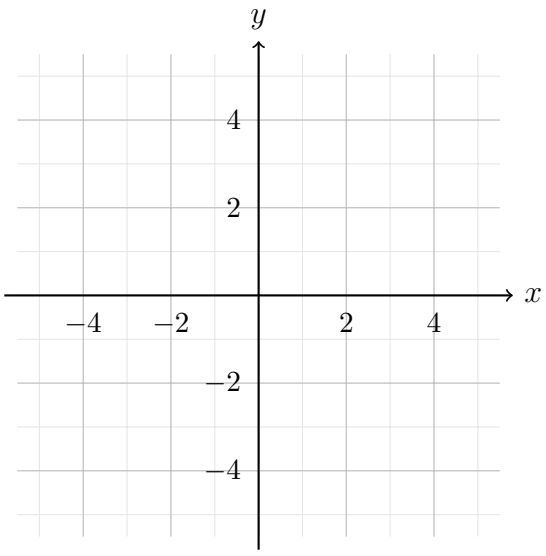
A. $y = 2$ **B.** $y = -2$ **C.** $y = 5$ **D.** $y = -5$

AM5. Which point rule describes a shift **down 4**?

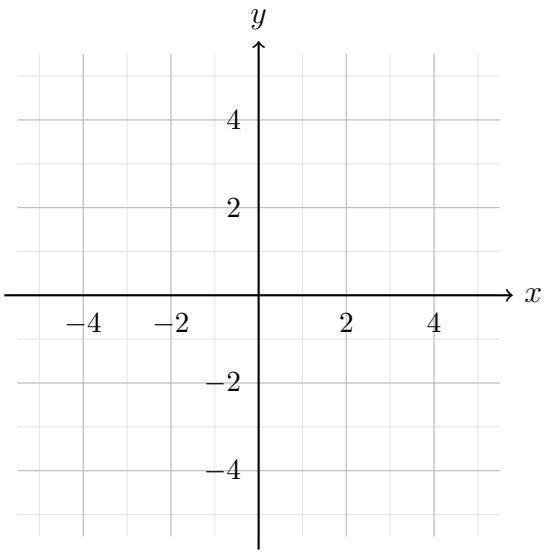
A. $(x, y) \to (x, y+4)$ **B.** $(x, y) \to (x-4, y)$ **C.** $(x, y) \to (x, y-4)$ **D.** $(x, y) \to (x + 4, y)$

Part A — Graph It!

AG1. On the axes below, plot at least 5 key points for $y = x^2$ (**blue**), then sketch $y = x^2 - 3$ (**red**) on the same axes. Label each curve and mark the vertex of each.



AG2. On the axes below, sketch both $y = 2^x$ (**blue**) and $y = 2^{x+2}$ (**red**). Plot at least 4 points per curve. Label the y-intercept of each. State the horizontal asymptote.

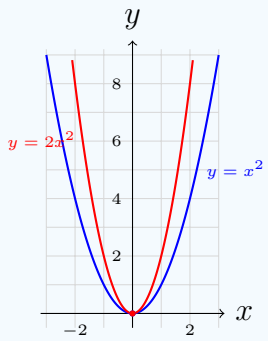


Asymptote: _____

Part B — Dilations

Worked Example: Transform $y = x^2$ with a **vertical stretch by 2**: $y = 2x^2$.

Parent $y = x^2$		**New** $y = 2x^2$
x	y	y
-2	4	8
-1	1	2
0	0	0
1	1	2
2	4	8



Point rule: $(x, y) \to (x,\, 2y)$
Same vertex $(0, 0)$; graph is *narrower*.

B1. Describe the transformation from $y = x^2$ to $y = 5x^2$.

B2. Describe the transformation from $y = x^2$ to $y = \dfrac{1}{3}x^2$.

B3. Describe the transformation from $y = 2^x$ to $y = 4 \cdot 2^x$.

B4. Describe the transformation from $y = 2^x$ to $y = \dfrac{1}{4} \cdot 2^x$.

B5. The point $(2, 4)$ lies on $y = x^2$. Find the corresponding point on $y = 3x^2$.

B6. The point $(1, 2)$ lies on $y = 2^x$. Find the corresponding point on $y = \dfrac{1}{2} \cdot 2^x$.

B7. Describe the transformation from $y = x^2$ to $y = (3x)^2$.

B8. Describe the transformation from $y = 2^x$ to $y = 2^{2x}$.

B9. Write the equation for $y = x^2$ after a **vertical stretch by 6**.

B10. The point $(4, 16)$ lies on $y = x^2$. After a horizontal compression by $\frac{1}{2}$, what is the new point?

Part B — Multiple Choice (circle your answer)

BM1. Which equation is a **vertical stretch** of $y = x^2$ by a factor of 4?

A. $y = \frac{x^2}{4}$ **B.** $y = (4x)^2$ **C.** $y = 4x^2$ **D.** $y = \left(\frac{x}{4}\right)^2$

BM2. The point $(3, 9)$ on $y = x^2$ maps to which point on $y = \frac{1}{3}x^2$?

A. $(3, 3)$ **B.** $(1, 9)$ **C.** $(9, 9)$ **D.** $(3, 27)$

BM3. Which equation is a **horizontal stretch** of $y = 2^x$ by a factor of 2?

A. $y = 2^{2x}$ **B.** $y = 2 \cdot 2^x$ **C.** $y = 2^{x/2}$ **D.** $y = \frac{1}{2} \cdot 2^x$

BM4. After a vertical compression by $\frac{1}{5}$, the point $(2, 4)$ on $y = x^2$ becomes:

A. $(10, 4)$ **B.** $(2, 20)$ **C.** $\left(2, \frac{4}{5}\right)$ **D.** $\left(\frac{2}{5}, 4\right)$

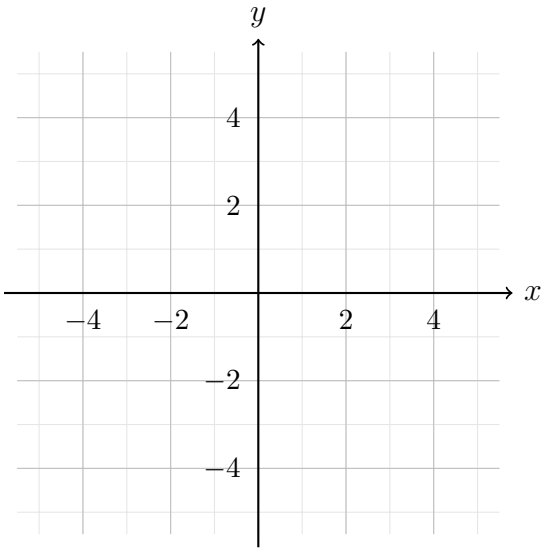
BM5. The transformation $y = \left(\frac{1}{2}x\right)^2$ is best described as:

A. Vertical compression by $\frac{1}{2}$ **B.** Horizontal stretch by 2

C. Vertical stretch by 2 **D.** Horizontal compression by $\frac{1}{2}$

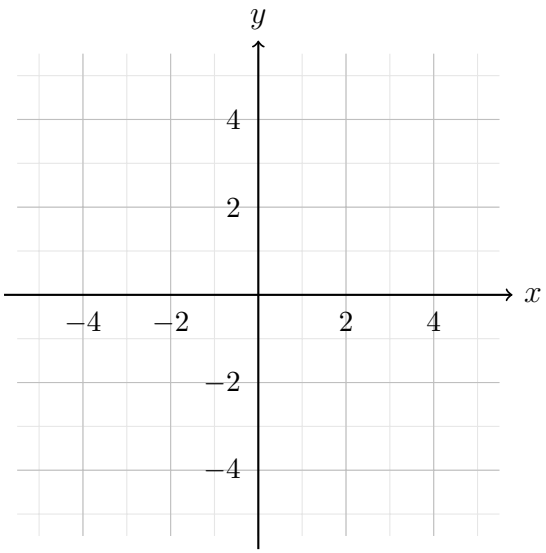
Part B — Graph It!

BG1. On the axes below, plot $y = x^2$ (**blue**) and $y = \frac{1}{2}x^2$ (**red**) using at least 5 key points each. How do the two graphs compare in width?



Comparison: _____

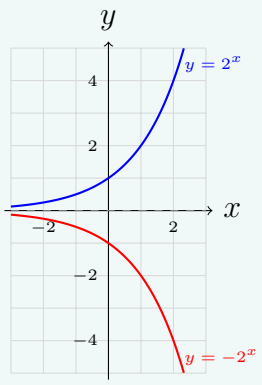
BG2. On the axes below, sketch $y = 2^x$ (**blue**) and $y = 3 \cdot 2^x$ (**red**) using at least 4 points each. Label the y-intercept of each curve.



Part C — Reflections

Worked Example: Reflect $y = 2^x$ over the x-axis to get $y = -2^x$.

Parent $y = 2^x$		**New** $y = -2^x$
x	y	y
-2	$\frac{1}{4}$	$-\frac{1}{4}$
-1	$\frac{1}{2}$	$-\frac{1}{2}$
0	1	-1
1	2	-2
2	4	-4



Point rule: $(x, y) \to (x, -y)$
Asymptote stays $y = 0$; graph flips down.

C1. Describe the transformation from $y = x^2$ to $y = -x^2$.

C2. Is $y = (-x)^2$ the same as $y = x^2$? Explain.

C3. Describe the transformation from $y = 2^x$ to $y = -2^x$.

C4. Describe $y = 2^{-x}$. Is this function increasing or decreasing?

C5. The point $(2, 4)$ lies on $y = x^2$. Find the corresponding point on $y = -x^2$.

C6. The point $(1, 2)$ lies on $y = 2^x$. Find the corresponding point on $y = 2^{-x}$.

C7. The point $(-3, 9)$ lies on $y = x^2$. Find its image after reflecting over the x-axis.

C8. Write the equation for $y = 2^x$ reflected over the y-axis.

C9. Write the equation for $y = x^2 - 3$ reflected over the x-axis. Simplify.

C10. List **all** transformations applied to $y = x^2$ to obtain $y = -(x + 2)^2 + 1$. State the correct order.

Part C — Multiple Choice (circle your answer)

CM1. The reflection of the point $(3, 9)$ over the x-axis is:

A. $(-3, 9)$ **B.** $(3, -9)$ **C.** $(-3, -9)$ **D.** $(9, 3)$

CM2. Which equation reflects $y = x^2 - 4$ over the x-axis?

A. $y = -x^2 + 4$ **B.** $y = x^2 + 4$ **C.** $y = (-x)^2 - 4$ **D.** $y = -x^2 - 4$

CM3. After reflecting $y = 2^x$ over the y-axis, what is the new y-intercept?

A. $(0, -1)$ **B.** $(0, 0)$ **C.** $(0, 2)$ **D.** $(0, 1)$

CM4. Which equation reflects $y = 2^{x-1} + 3$ over the x-axis?

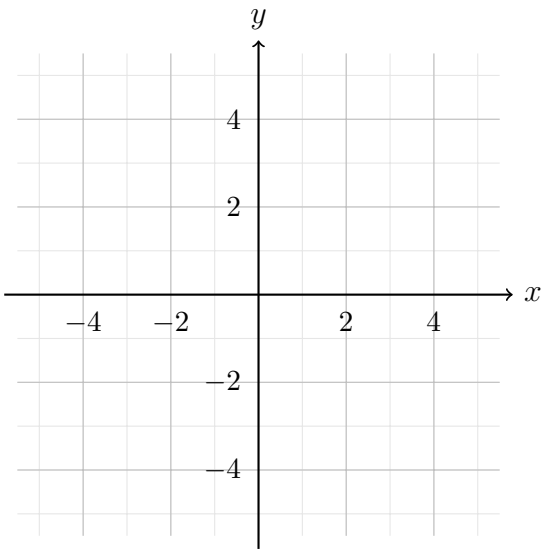
A. $y = -2^{x-1} + 3$ **B.** $y = -2^{x-1} - 3$ **C.** $y = 2^{-(x-1)} + 3$ **D.** $y = 2^{-x-1} + 3$

CM5. Which statement about $y = (-x)^2$ is true?

A. It equals $-x^2$ **B.** It equals x^2 **C.** It opens downward **D.** Its vertex is $(-1, 0)$

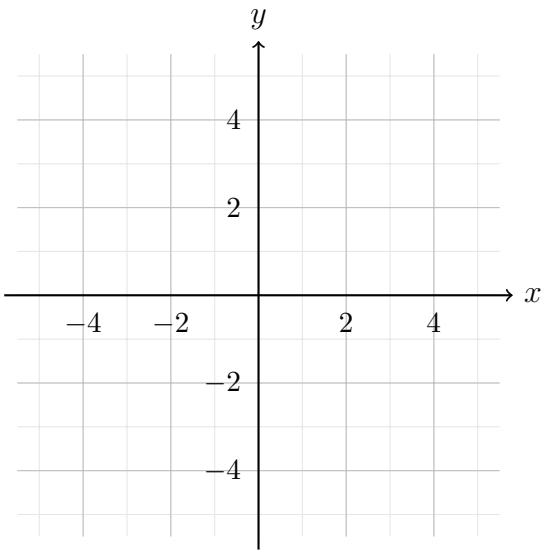
Part C — Graph It!

CG1. On the axes below, graph $y = x^2$ (**blue**) and $y = -x^2$ (**red**) using at least 5 key points each. Describe the relationship between the two curves.



Relationship: _____

CG2. On the axes below, sketch $y = 2^x$ (**blue**) and $y = 2^{-x}$ (**red**). Plot at least 4 points per curve. What is the same about both graphs? What is different?



Same: _____ *Different:* _____

Part D — Combined Transformations (Multiple Choice)

D1. Which equation represents $y = x^2$ shifted **right 3**, vertically **stretched by 2**, and shifted **up 1**?

A. $y = 2(x+3)^2 + 1$ **B.** $y = 2(x-3)^2 + 1$ **C.** $y = 2(x-3)^2 - 1$ **D.** $y = \frac{1}{2}(x-3)^2 + 1$

D2. The point $(1, 2)$ on $y = 2^x$, after reflecting over the x-axis then shifting **up 3**, becomes:

A. $(1, 1)$ **B.** $(-1, 5)$ **C.** $(1, 5)$ **D.** $(1, -5)$

D3. Which equation is a **vertical compression** of $y = x^2$ by $\frac{1}{4}$?

A. $y = 4x^2$ **B.** $y = (4x)^2$ **C.** $y = \frac{1}{4}x^2$ **D.** $y = \left(\frac{1}{4}x\right)^2$

D4. The horizontal asymptote of $y = 3 \cdot 2^{x+2} - 5$ is:

A. $y = 3$ **B.** $y = 2$ **C.** $y = -5$ **D.** $y = 5$

D5. Which correctly describes all transformations of $y = -2(x-1)^2 + 4$?

A. Shift left 1, stretch 2, reflect x-axis, up 4 **B.** Shift right 1, stretch 2, reflect x-axis, up 4

C. Shift right 1, compress 2, reflect x-axis, down 4 **D.** Shift right 1, stretch 2, up 4

D6. The point $(3, 8)$ on $y = 2^x$, after the transformation $y = -2^x + 5$, becomes:

A. $(3, -3)$ **B.** $(-3, -3)$ **C.** $(3, 13)$ **D.** $(3, 3)$

D7. The vertex of $y = -3(x+2)^2 - 1$ is:

A. $(2, -1)$ **B.** $(-2, -1)$ **C.** $(2, 1)$ **D.** $(-2, 1)$

D8. Which equation represents $y = 2^x$ reflected over the y-axis, then shifted up 3?

A. $y = -2^x + 3$ **B.** $y = 2^{-x} - 3$ **C.** $y = 2^{-x} + 3$ **D.** $y = -2^{-x} + 3$

D9. The point $(6, 36)$ on $y = x^2$, after a horizontal compression by $\frac{1}{3}$, becomes:

A. $(6, 12)$ **B.** $(18, 36)$ **C.** $(2, 36)$ **D.** $(2, 12)$

D10. Which equation has vertex $(-1, 5)$, opens **downward**, and is **narrower** than $y = x^2$?

A. $y = 2(x-1)^2 + 5$ **B.** $y = -2(x+1)^2 + 5$ **C.** $y = -\frac{1}{2}(x+1)^2 + 5$ **D.** $y = 2(x+1)^2 - 5$

Answer Key

Part A — Translations

A1. Shift **up 5 units**. $(x, y) \to (x, y + 5)$.

A2. Shift **right 3 units**. $(x, y) \to (x + 3, y)$.

A3. Shift **down 4 units**. Asymptote moves to $y = -4$.

A4. Shift **left 1 unit**. Asymptote stays $y = 0$.

A5. $y = (x + 2)^2 - 3$ (vertex at $(-2, -3)$).

A6. $y = 2^{x-5} + 2$ (asymptote: $y = 2$; y-int: $(0, \frac{1}{32} + 2)$).

A7. $(3, 16)$ — y increases by 7: $9 + 7 = 16$.

A8. $(3, 4)$ — x shifts right 1: $x = 3$, $y = (3 - 1)^2 = 4$.

A9. Vertex: $(4, 3)$.

A10. Asymptote: $y = -6$.

Part A MC: AM1: **B** AM2: **B** AM3: **C** AM4: **C** AM5: **C**
Part A Graphing:

- **AG1:** $y = x^2$ vertex $(0, 0)$; $y = x^2 - 3$ vertex $(0, -3)$, same shape, shifted down 3.
- **AG2:** $y = 2^x$ has y-int $(0, 1)$; $y = 2^{x+2}$ has y-int $(0, 4)$ (shifted left 2). Both: asymptote $y = 0$.

Part B — Dilations

B1. Vertical **stretch** by factor 5 (graph narrower). $(x, y) \to (x, 5y)$.

B2. Vertical **compression** by factor $\frac{1}{3}$ (graph wider).

B3. Vertical **stretch** by factor 4. y-int moves from $(0, 1)$ to $(0, 4)$.

B4. Vertical **compression** by factor $\frac{1}{4}$. y-int: $(0, \frac{1}{4})$.

B5. $(2, 12)$ y-value $\times 3$: $4 \times 3 = 12$.

B6. $(1, 1)$ y-value $\times \frac{1}{2}$: $2 \times \frac{1}{2} = 1$.

B7. Horizontal **compression** by $\frac{1}{3}$ (same as vertical stretch by 9, since $(3x)^2 = 9x^2$).

B8. Horizontal **compression** by $\frac{1}{2}$. $(x, y) \to (\frac{x}{2}, y)$.

B9. $y = 6x^2$.

B10. $(2, 16)$ $x \div 2 = 2$, y unchanged.

Part B MC: BM1: **C** BM2: **A** BM3: **C** BM4: **C** BM5: **B**
Part B Graphing:

- **BG1:** $y = x^2$ is *wider*; $y = \frac{1}{2}x^2$ is *wider* than $y = x^2$. Both share vertex $(0, 0)$.
- **BG2:** $y = 2^x$: y-int $(0, 1)$; $y = 3 \cdot 2^x$: y-int $(0, 3)$. Second is a vertical stretch; grows faster.

Part C — Reflections

C1. Reflection over the x**-axis**; graph opens downward. $(x, y) \to (x, -y)$.

C2. Yes, same graph. $(-x)^2 = x^2$ — parabola is an even function.

C3. Reflection over the x**-axis**; graph is below the x-axis; asymptote stays $y = 0$.

C4. Reflection over the y**-axis**. The function is **decreasing** (as x increases, 2^{-x} decreases).

C5. $(2, -4)$ negate y: $4 \to -4$.

C6. $(-1, 2)$ negate x: $1 \to -1$.

C7. $(-3, -9)$ on $y = -x^2$.

C8. $y = 2^{-x}$.

C9. $y = -(x^2 - 3) = -x^2 + 3$.

C10. Order: **(1)** shift left 2, **(2)** reflect over x-axis, **(3)** shift up 1.

Part C MC: CM1: **B** CM2: **A** CM3: **D** CM4: **B** CM5: **B**
Part C Graphing:

- **CG1:** $y = -x^2$ is a mirror image of $y = x^2$ reflected over the x-axis. Same shape, opens down.
- **CG2:** Same: both have y-int $(0, 1)$ and asymptote $y = 0$. Different: $y = 2^x$ increases, $y = 2^{-x}$ decreases. They are mirror images over the y-axis.

Part D — Combined MC

D1: B D2: A D3: C D4: C D5: B D6: A D7: B D8: C D9: C D10: B

Selected explanations:

- D2: Reflect $(1, 2) \to (1, -2)$; shift up 3: $-2 + 3 = 1$. New point $(1, 1)$.
- D4: Only the -5 outside shifts the asymptote from $y = 0$ to $y = -5$.
- D6: $-2^3 + 5 = -8 + 5 = -3$. New point $(3, -3)$.
- D9: Horiz. compression by $\frac{1}{3}$: $x \to \frac{x}{3} = \frac{6}{3} = 2$, y unchanged. Point $(2, 36)$.